

DRY-RUN GUIDE | Binar_search.py
Source: Dsa\search\Binar_search.py

CORE BEHAVIOR

Binary-searches an already sorted sequence by discarding half the remaining interval after each comparison.

REPRESENTATIVE INPUT

[1,3,5,7,9], target 7.

STEP TRACE

mid=2 value 5 -> search right; mid=3 value 7 -> found.

EXPECTED OUTPUT / RETURN

Index 3 / source-formatted found result.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\search\Binar_search.py; line numbers refer to the original file. No source was modified.

```
001 | def binary_search(arr, target):
002 |     """
003 |     Performs Iterative Binary Search on a sorted list.
004 |     Returns the index if found, else returns -1.
005 |     """
006 |     low = 0
007 |     high = len(arr) - 1
008 |
009 |     while low <= high:
010 |         # Prevents potential overflow in lower-level languages
011 |         mid = low + (high - low) // 2
012 |
013 |         if arr[mid] == target:
014 |             return mid
015 |         elif arr[mid] < target:
016 |             low = mid + 1 # Search right half
017 |         else:
018 |             high = mid - 1 # Search left half
019 |
020 |     return -1
021 |
022 |
023 | # Example Usage:
024 | sorted_arr = [5, 10, 15, 20, 25, 30, 35]
025 | target = 30
026 | result = binary_search(sorted_arr, target)
027 |
028 | if result != -1:
029 |     print(f"Element found at index {result}")
030 | else:
031 |     print("Element not found")
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.